Quiz 2 formula sheet

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CH 7 RL switching · find current and voltage

Before and after the switch

Draw each switch position. At DC steady state short L and open C. Solve the old circuit for I_0 and V_0, then the new circuit for the final values.

$$i_L(0^+) = i_L(0^-) = I_0, \quad v_C(0^+) = v_C(0^-) = V_0$$

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}, \qquad \tau = \frac{L}{R_{\text{Th}}}$$

Find RTh looking into the inductor terminals after switching. Short independent voltage sources and open independent current sources. Trace the discharge path.

Source removed or still connected?

$$i_L(t) = i_L(\infty) + [I_0 - i_L(\infty)]e^{-t/\tau}$$

Source-free: $i_L(t) = I_0 e^{-t/\tau}$

$$v_L = L\, di_L/dt, \qquad v_R = Ri_R$$

Series DC: $i_L(\infty) = V_s/R, \quad v_L = V_s - Ri_L$

Current enters the + terminal. At a later switch restart with the current at that instant and elapsed time. At steady DC, vL = 0.

CH 8 Switched RLC · find the response and both constants

Reduce the post-switch circuit

$$\alpha_{\text{series}} = \frac{R}{2L}, \quad \alpha_{\text{parallel}} = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}, \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Use equivalent R in the reduced loop or parallel branches. For critical R or C, set α = ω_0.

Find the initial slope from the diagram

$$i_L'(0^+) = v_L(0^+)/L, \quad v_C'(0^+) = i_C(0^+)/C$$

Series loop: use KVL with I_0 and V_0. If the capacitor drives the loop, Li′(0⁺) = V_0 − RI_0. Use the drawn polarity.

Source-free parallel, all branch currents downward: Cv′(0⁺) = −V_0/R − I_0. With a source injecting current, add its current to the right side.

Set the starting value for the transient

$$y(t) = x(t) - x(\infty), \quad y_0 = x(0^+) - x(\infty),$$

$$d = x'(0^+)$$

Choose the requested x (i or v). Source-free: x(∞) = 0. For a series voltage step, vC(∞) = Vs. For a parallel current step, iL(∞) = Is. Add x(∞) back to y(t).

Overdamped · α > ω_0

$$y = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$A_1 + A_2 = y_0, \qquad s_1 A_1 + s_2 A_2 = d$$

$$A_1 = \frac{d - s_2 y_0}{s_1 - s_2}, \qquad A_2 = y_0 - A_1$$

Underdamped · α < ω_0

$$y = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$A_1 = y_0, \qquad A_2 = (d + \alpha A_1)/\omega_d$$

Her source-free series current uses B_1, B_2 for these same cosine and sine coefficients.

Critically damped · α = ω_0

$$y = (A_1 + A_2 t)e^{-\alpha t}, \quad A_1 = y_0, \quad A_2 = d + \alpha y_0$$

Above: parallel response and the step-response forms. Her source-free series current reverses the labels:

$$i = (A_2 + A_1 t)e^{-\alpha t}, \quad A_2 = I_0, \quad A_1 = d + \alpha I_0$$

CH 9 AC voltage and current · cosine phasors

Convert the source and the components

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow \mathbf{V} = V_m \angle \phi$$

$$\sin\theta = \cos(\theta - 90^\circ), \quad \omega = 2\pi f, \quad T = 1/f$$

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = 1/(j\omega C)$$

Keep peak magnitudes. Add impedances in series. For PS8's series RLC circuit, X = ωL − 1/(ωC).

Solve for the requested current or voltage

Series RL: $Z = R + j\omega L, \qquad \mathbf{I} = \mathbf{V}_s/Z$

$$\mathbf{V}_R = R\mathbf{I}_R, \quad \mathbf{V}_L = j\omega L \mathbf{I}_L, \quad Z = \mathbf{V}/\mathbf{I}$$

Parallel RL: $\mathbf{I}_s = \mathbf{V}/R + \mathbf{V}/(j\omega L)$

For Z = R + jX, |Z| = √(R² + X²) and θ = atan2(X,R). Divide magnitudes, subtract angles. $I_m \angle \beta$ gives i(t) = I_m cos(ωt + β).

PS 6–8 Related problem-set formulas

DC switching and stored energy

$$w_L = \tfrac{1}{2} L i_L^2, \quad w_R = \int_{t_1}^{t_2} R i_R^2 \, dt$$

$$\tau_{RC} = R_{\text{Th}} C, \quad v_C(t) = V_f + (V_0 - V_f)e^{-t/\tau_{RC}}$$

Mixed AC branches

$$Z_p = \frac{Z_1 Z_2}{Z_1 + Z_2}, \quad \mathbf{V}_x = \mathbf{V}_s \frac{Z_x}{\sum Z_{\text{series}}}$$

$$\mathbf{I}_1 = \mathbf{I}_s \frac{Z_2}{Z_1 + Z_2} \quad \text{(two parallel branches)}$$